Geochemistry, Geophysics, Geosystems

Supporting Information for

Dynamic plume-ridge interaction at near-ridge hotspots: Evidence from new Galápagos gravity data

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Figures S1, S2, and S3.

Introduction

Supporting information includes two figures and captions. Figure S1 contains compiled data from Jefferson et al. (2014) and creates a data-driven visual representation of the abrupt absence of calderas in the Galapagos. Figure S2 contains the graphic representation of the flexure calculation for the Galapagos platform and its effect on the surveys in the study. Figure S3 contains a plot of the components of the Complete Bouguer Anomaly from each the submarine and above sea level masses. Table S1 contains bin numbers for sample reference.

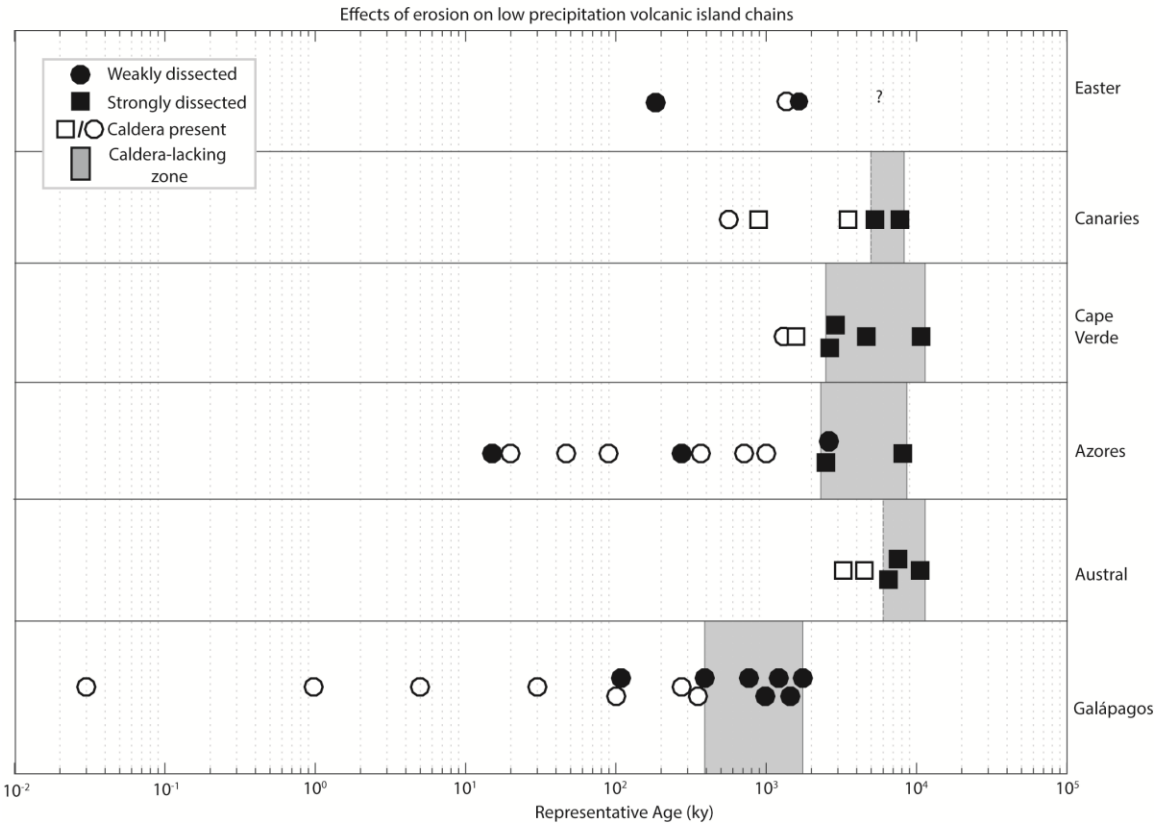
Effects of erosion on low precipitation volcanic island chains
Weakly dissected
Strongly dissected
Caldera present
Caldera-lacking zone
?
Easter
Canaries
Cape Verde
Azores
Austral
Galápagos
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
Representative Age (ky)

Figure S1. Dissection and caldera presence versus representative age for volcanoes with low precipitation rates. Representative age is reported on the basis of: the average or shield age, the median of the oldest and youngest age available, or the reported age if only one age is available (Jefferson et al., 2014). The grey box highlights the age range for which calderas are absent in each island chain. Eastern Islands lack this zone because there are not enough data points to demarcate a distinction between caldera present and caldera absent. Some markers are shifted vertically to accommodate overlap. The Galápagos caldera-free zone is shifted earlier compared to other island chains with similarly low precipitation rates, indicating that caldera obfuscation is occurring earlier in island chain evolution than other island chains."

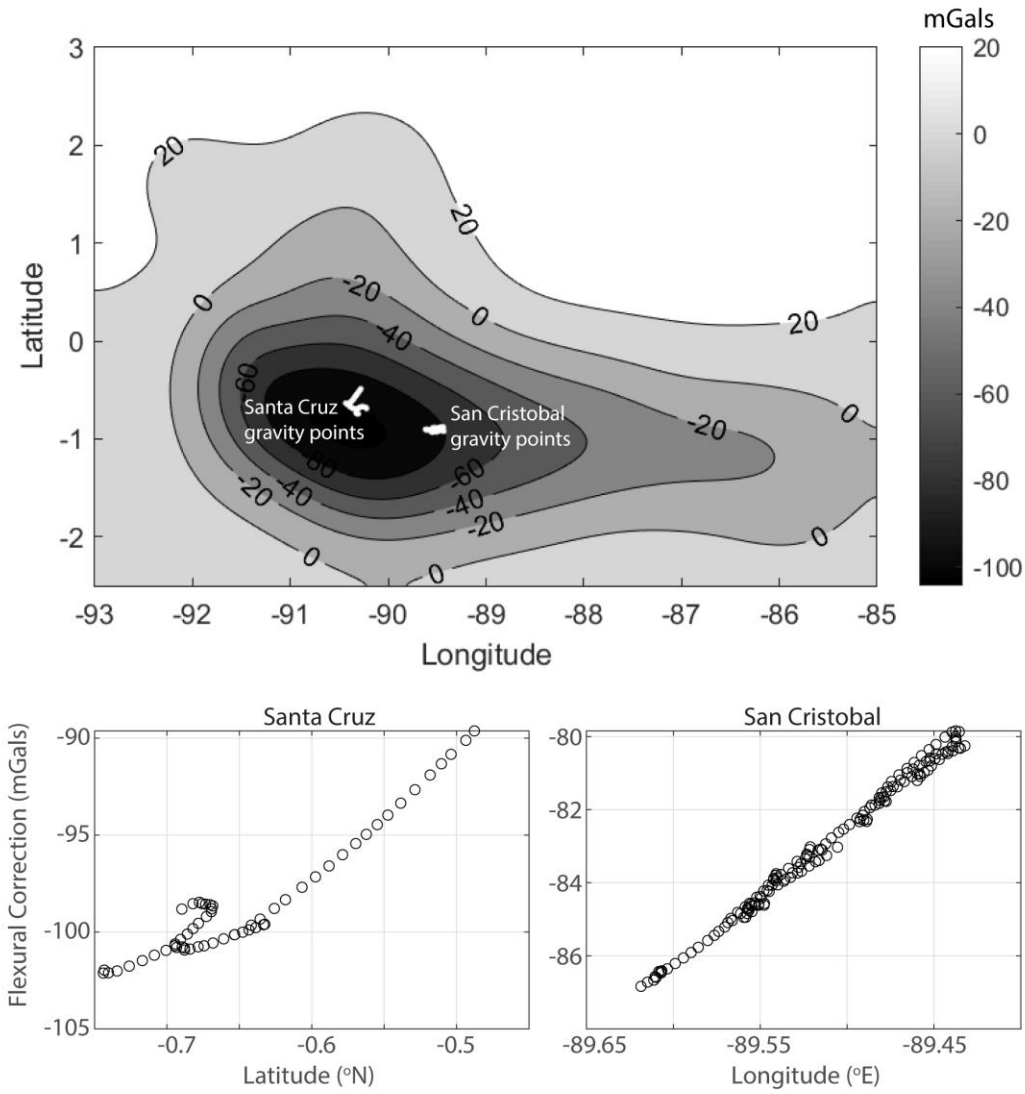

Figure S2. Flexure calculations for the Galápagos platform. (top) Contours of flexural deflection of a constant thickness plate with rigidity of 10^{22} N/m (Karnauskas et al., 2017) showing the archipelago-wide flexure profile. (bottom) The gravitational effect of this flexure does not act to emphasize any local gravitational features on the survey data points for Santa Cruz (left) and San Cristobal (right).

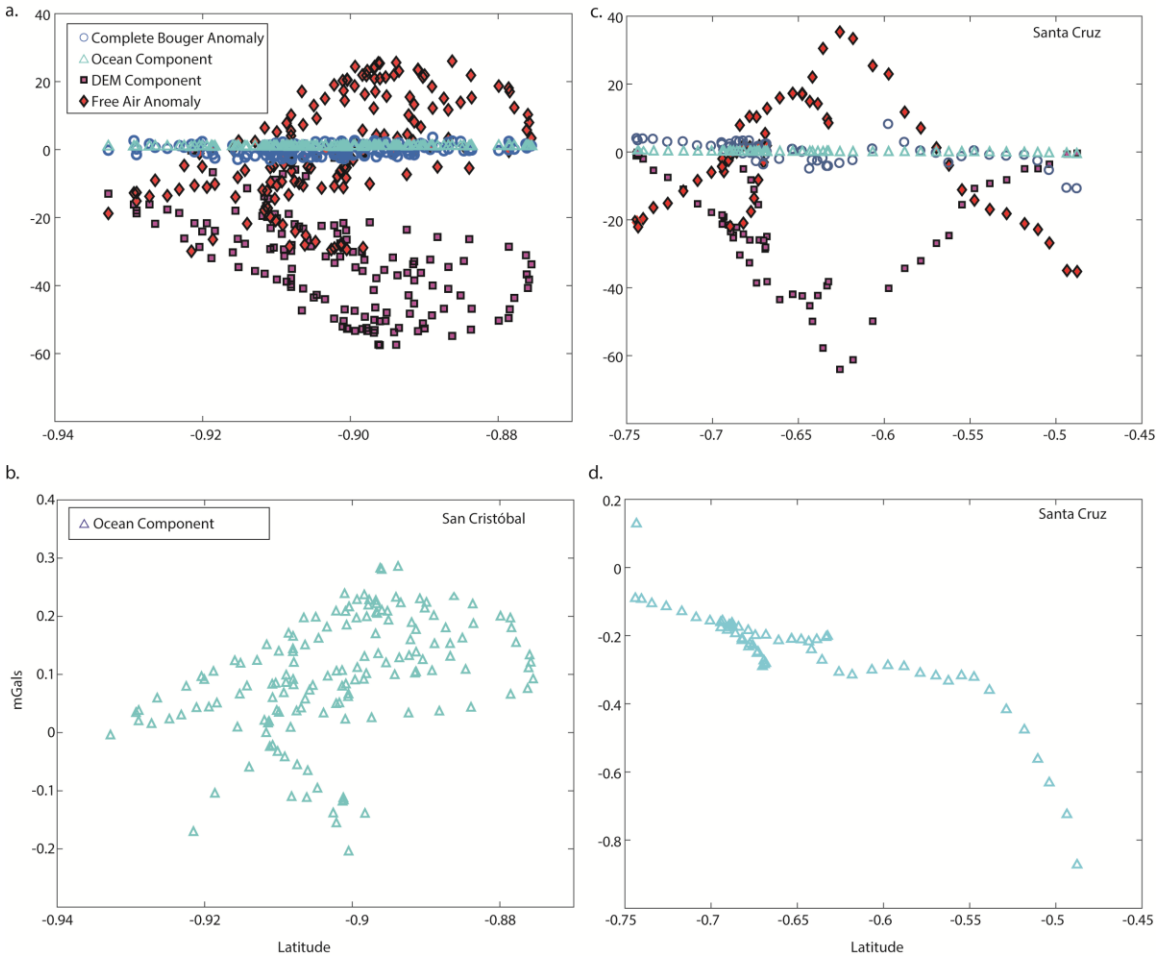

Figure S3. Components of the Complete Bouguer Anomaly (BA). (a) and (c) display the San Cristobal and Santa Cruz BA components, respectively, that is contributed from the digital elevation map (DEM; filled magenta squares), the component contributed from "filling-in" of the ocean (open cyan triangles), the Complete BA minus the mean of the Complete BA (open blue circles), and the Free Air Anomaly minus the mean of the Free Air Anomaly (filled red diamonds). (b) and (d) display a more detailed view of the ocean component contributing to the San Cristobal and Santa Cruz Complete BA from (a) and (c), respectively."

Volcano	Number	Reference
Cerro Azul Volcano, Isabela	80	Teasdale et al., 2005; Saal et al., 2007
Ecuador Volcano, Isabela	292	Geist et al., 2002
Fernandina	69	Allan and Simkin, 2000; Saal et al., 2007
Wolf Volcano, Isabela	190	Geist et al., 2005
Darwin Volcano, Isabela	2	Saal et al., 2007
Sierra Negra, Isabela	97	Geist et al., 2008; Reynolds and Geist, 1996
Alcedo Volcano, Isabela	44	Geist et al., 1995
Santiago	32	Saal et al., 2007; Gibson and Geist, 2010; Gibson et al., 2012
Santa Cruz	105	Wilson, 2013; Saal et al., 2007
Espanola	26	McGuire et al., 2015
San Cristóbal	78	Pimentel et al., 2016; Mahr et al., 2016

Table S1. Sources of Mg# values for Galápagos volcanoes. Number refers to number of samples from the corresponding references.